

Architecture: AI Competitor Intelligence Pipeline

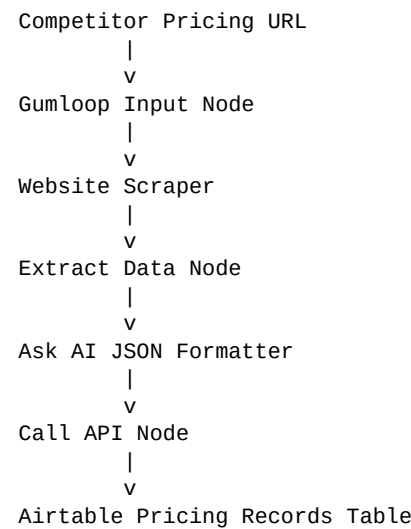
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1. Architecture Overview

The **AI Competitor Intelligence Pipeline** is built as a visual workflow in Gumloop. The system takes a competitor pricing page URL, scrapes page content, extracts structured pricing information using AI, formats the result into Airtable API JSON, and sends the records into Airtable.

Webhook.site was used as a temporary testing endpoint before connecting the production destination, Airtable.

2. High-Level Flow



3. Components

3.1 Gumloop Input Node

The input node accepts the competitor pricing page URL. This makes the workflow reusable and avoids hardcoding a single pricing page.

Item	Value
Input field	competitor_url
Example value	https://www.notion.com/pricing
Purpose	Allow different competitor URLs to be tested.

3.2 Website Scraper Node

The Website Scraper node visits the competitor pricing URL and extracts page content as text.

Input	Output	Purpose
competitor_url	Website Content	Converts the live pricing page into raw text for AI extraction.

3.3 Extract Data Node

The Extract Data node uses AI to convert messy scraped website content into structured fields. Pricing pages usually contain multiple plans, so the node returns list outputs.

Extracted Output
Competitor Name List
Plan Name List
Price List
Billing Period List
Target User List
Key Features List
PM Insight List

3.4 Ask AI Node

The Ask AI node formats each extracted pricing plan into an Airtable-compatible JSON body. Loop Mode is ON because Extract Data returns lists. Ask AI processes one pricing plan at a time and creates one JSON object per plan.

Setting	Value
Loop Mode	ON
Input	Extract Data list outputs
Output	Airtable-ready JSON response list
Reason	One JSON object is generated per pricing plan.

Expected output format:

```
{
  "fields": {
    "Competitor Name": "Notion",
    "Plan Name": "Business",
    "Price": "$20",
    "Billing Period": "per member / month",
    "Target User": "growing businesses",
    "Key Features": "SAML SSO, private teamspaces, advanced permissions",
    "PM Insight": "Business plan targets growing teams that need admin and security controls.",
    "Source URL": "https://www.notion.com/pricing",
    "Date Checked": "2026-07-02"
  }
}
```

Important: Airtable field names must match the table columns exactly. For example, use **Competitor Name**, not **competitor_name**.

3.5 Call API Node

The Call API node sends each JSON record to Airtable using a POST request. Loop Mode is ON because the Ask AI node creates one JSON response per pricing plan.

Setting	Value
Method	POST
Loop Mode	ON
Body	Ask AI Response List
Header 1	Authorization: Bearer <AIRTABLE_PERSONAL_ACCESS_TOKEN>

Setting	Value
Header 2	Content-Type: application/json

Airtable endpoint format:

`https://api.airtable.com/v0/<BASE_ID>/<TABLE_NAME_OR_TABLE_ID>`

Example using a table name:

`https://api.airtable.com/v0/appXXXXXXXXXXXX/Pricing%20Records`

Example using a table ID:

`https://api.airtable.com/v0/appXXXXXXXXXXXX/tblXXXXXXXXXXXX`

4. Airtable Destination

Airtable acts as the final storage layer. The table is named **Pricing Records**, and each row represents one competitor pricing plan.

Competitor Name	Plan Name	Price	Billing Period
Notion	Free	\$0	per member / month
Notion	Plus	\$10	per member / month
Notion	Business	\$20	per member / month
Notion	Enterprise	Custom pricing	per member / month

5. Webhook.site Testing Layer

Webhook.site was used before Airtable to validate the API request. It was a temporary debugging destination, not the final storage system.

- Confirmed Gumloop was sending POST requests.
- Allowed inspection of raw JSON payloads.
- Validated Loop Mode behavior.
- Helped debug payload formatting before Airtable authentication was added.

6. Data Flow

- 1 User provides a competitor URL through the Gumloop input node.
- 2 Website Scraper extracts raw pricing page text.
- 3 Extract Data creates structured list outputs for plans, prices, features, and insights.
- 4 Ask AI creates an Airtable JSON payload for each pricing plan.
- 5 Call API sends each JSON body to Airtable using POST.
- 6 Airtable creates one new row per pricing plan.

7. Loop Mode Design

Loop Mode is used because one competitor pricing page contains multiple pricing plans.

Node	Loop Mode	Reason
Extract Data	Returns lists	Pricing pages contain multiple plans.
Ask AI	ON	Processes one item from each list at a time.
Call API	ON	Sends one POST request per pricing plan.

Example loop behavior:

Loop 1 -> Free plan
 Loop 2 -> Plus plan
 Loop 3 -> Business plan
 Loop 4 -> Enterprise plan

8. Security Considerations

The workflow uses an Airtable Personal Access Token. This token and other private values must not be committed to GitHub or shown in public screenshots.

- Do not commit Airtable tokens to GitHub.
- Do not include private API URLs in public screenshots.
- Do not expose Webhook.site private URLs.
- Blur sensitive credentials in documentation.
- Store credentials only inside Gumloop or a secure environment.

Sensitive values:

Airtable Personal Access Token
 Airtable Base ID
 Airtable Table ID
 Webhook.site URL

9. Known Limitations

- Processes one competitor URL at a time.
- Does not automatically detect changes from previous runs.
- Does not remove duplicate Airtable rows.
- Does not schedule recurring runs yet.
- AI extraction may need adjustment for different pricing page layouts.
- Pages requiring login or heavy interaction may not scrape correctly.

10. Future Architecture Improvements

- Add multi-URL input.
- Add scheduled weekly execution.
- Add deduplication logic.
- Add change detection against previous records.
- Add Slack or email alerts.
- Add raw scraped text storage.
- Add confidence scoring for extracted fields.

- Add Airtable dashboard views.
- Add error handling and retry logic.

11. Final Architecture Summary

The project uses Gumloop as the automation layer, AI as the extraction and formatting layer, and Airtable as the structured database. The final architecture converts live competitor pricing pages into structured PM intelligence records that can be reviewed, compared, and expanded over time.